Maximum A Posteriori Inference for Wumpus Trajectory Tracking

Amy McDermott (26911264)

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I declare that this report and the work presented within it are my own. It has been developed with the support of the module coursework and provided lecture notes. While the mathematical derivations, algorithmic logic, and interpretation of results are my own work, Gemini was used to assist in achieving a professional document through technical LaTeX formatting. This includes help with the layout of figures and tables, the formatting of mathematical equations and code listings, as well as assistance in the implementation of the Python tool used for the animated visualisation of results.

ECSA graduate attributes

GA 1: Problem solving

This attribute was met through the formulation (Section 1 and 2), implementation (Section 3) and resolution (Section 4) of the wumpus tracking task. The problem was first analysed by identifying the complexities of the stochastic movement and unreliable detection data. Thereafter, the ability to research literature is demonstrated by incorporating Hidden Markov Model frameworks to model the latent trajectory (Section 2.1.2). Substantiated conclusions (Section 4) were reached through the implementation of a Probabilistic Graphical Model (PGM) solution from first principles (Section 2) as well as an EM algorithm (Section 3.6) to estimate unknown parameters (p_w, p_c), rather than relying on point estimates.

GA 2: Application of scientific and engineering knowledge

This GA was satisfied by applying knowledge of probability theory and PGMs to solve a latent entity tracking task. Specifically, the PGM formulation is described in Section 2.1.2, and detailed mathematical formulations, including the E-step responsibilities and M-step update equations, are documented in Section 2.3. Furthermore, the Sum-Product algorithm for marginal inference was used during the parameter estimation phase and thereafter the Max-Sum (MAP) algorithm for inference to obtain a globally consistent trajectory (Section 3.6). The application of these mathematical frameworks allowed for the successful tracking of a latent entity across different environmental contexts (Section 4).

1 Problem Description

This project addresses an object tracking task within a discrete grid world inhabited by a latent entity known as a wumpus. The objective is to determine the trajectory of the wumpus across multiple time steps based on inaccurate measurements. At each time step the wumpus attempts to move to an adjacent cell (up, down, left, or right) with equal probability. If the target cell is within the grid boundaries, the wumpus transitions to that location; otherwise it remains in its current position. The exact location of the wumpus is never directly observed. Instead, every cell in the grid generates a binary detection value at each time step, parameterised by:

- p_w (True Positive): The probability that the cell containing the wumpus correctly generates a detection.
- p_c (False Positive): The probability that a cell not containing the wumpus generates a detection (clutter measurement).

Given the stochastic grid detection data across multiple time steps, the task is to design, implement and evaluate a Probabilistic Graphical Model (PGM) solution. This involves constructing a model that accurately captures the wumpus's transition and observation logic to infer it's latent trajectory across various datasets.

2 Solution Design

The solution was developed iteratively to accomodate the increasing complexity and varying characteristics of each dataset. The PGM design process followed a structured methodology: defining the random variables, defining the model, and constructing the model factors. This process is detailed below for each attempted dataset.

2.1 Dataset 1

The initial model was designed for a 5×5 grid over a series of $T = 10$ time steps. In this case, the parameters $p_w = 0.95$ and $p_c = 0.05$ are known constants.

2.1.1 Random Variable Definition

A critical design choice was made to represent the wumpus's location using a single latent RV, W^t, instead of seperate X^t and Y^t coordinates. The grid is mapped to a 1-dimensional vector where each cell index $i \in \{0, \ldots, 24\}$ corresponds to a unique (x, y) coordinate. This approach offers several advantages. Primarily, it enhances inference efficiency by reducing the total number of RVs within the cluster graph, which leads to more efficient message passing and overall faster convergence. Furthermore, it reduces complexity by simplifying factor construction and avoiding the coordination of the dependencies between spatial coordinates. The wumpus's movement constraints are more easily enforced within a transition matrix; for example, the probability of illegal moves, such as diagonal steps, can be explicitly set to zero within one factor. The RVs are defined as follows:

- $W^t \in \{0, 1, \ldots 24\}$: The latent position of the wumpus at time t.
- D_i^t in {0,1}: The binary detection status of cell i at time t, where 1 indicates a detection.

2.1.2 Model Definition

The wumpus tracking task is formulated as a Hidden Markov Model (HMM), a probabilistic framework designed to infer a sequence of latent states from a series of observable outputs [1]. The trajectory of the wumpus $(W^0, W^1, \ldots, W^T)$ is modeled as a first-order Markov process, defined by the transition probability $P(W^t \mid W^{t-1})$. This assumes the memoryless Markov property, where the future state is independent of the past, given the current state. At each time step, the observable detection variables are treated as emissions generated from the current latent state, following the conditional distribution $P(\mathbf{D}^t \mid W^t)$. As a specific subclass of Bayesian Networks that is optimised for modeling sequential data, the HMM structure effectively captures the temporal dependencies of the wumpus's movement, as illustrated in the BN in Figure 1.

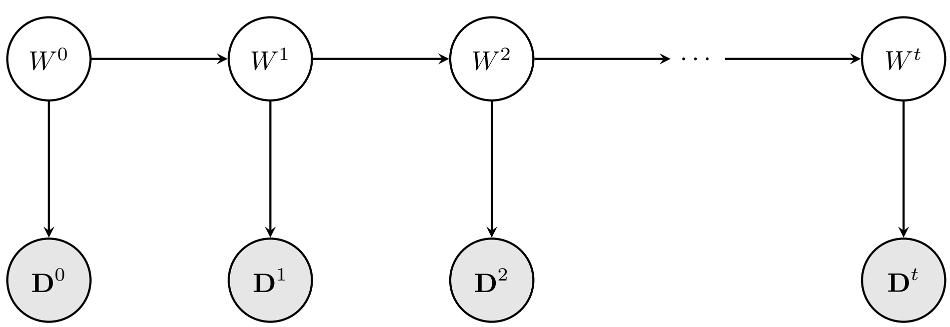


Figure 1: Bayesian Network structure of the HMM tracking solution.

2.1.3 Factor Definition

The model is governed by two primary factors. The first factor models the transition model of the wumpus over time as a first-order Markov process $P(W^t \mid W^{t-1})$. This factor captures the entity's movement across adjacent cells, where the wumpus attempts to move up, down, left, or right with an equal probability of 0.25 per direction. To enforce the physical boundaries of the grid, the wumpus remains in its current position if it attempts to move outside the grid world. Consequently, the probability of the wumpus remaining stationary ($P(W^t = i \mid W^{t-1} = i)$) is determined by its proximity to the boundaries:

- Interior cells: The wumpus always moves to an adjacent cell with a probability of 0.25. Thus, the probability of remaining in place is 0.
- Edge cells: With the wumpus having one direction blocked, the probability of remaining in place is 0.25.
- Corner cells: With the wumpus having two blocked directions, the probability of remaining in place increases to 0.5.

For a 5×5 grid, the stochastic wumpus trajectory results in a sparse 25×25 transition matrix that ensures that the inferred wumpus movement adheres to the constraints of the environment. This transition factor is invariant across all time steps $t \in \{1, \dots, T\}$ since the wumpus's rules of movement do not change as time progresses.

Table 1: Illustrative sample of the Transition Probability Factor $P(W^t \mid W^{t-1})$ for a 5×5 grid.

W^{t-1}	W^t	$P(W^t \mid W^{t-1})$	**Description**
0	0	0.50	Stay (Blocked Up and Left)
0	1	0.25	Valid Move (Right)
0	5	0.25	Valid Move (Down)
0	6	0.00	Illegal Move (Diagonal)
12	12	0.00	Stay (No Boundaries Hit)
12	13	0.25	Valid Move (Right)
...	...	...	...

Secondly, a factor $P(\mathbf{D}^t \mid W^t)$ that links the latent state W^t to the observed grid-wide detection data $\mathbf{D}^t$ is designed. This factor is parameterised by the detection probability p_w and the clutter probability p_c, which accounts for sensor inaccuracies across all cells in the grid. In the context of HMMs, this factor defines the likelihood of a specific observation being generated from a particular hidden state, otherwise known as an emission probability.

Table 2: Illustrative sample of the Detection Factor for a single cell i at time t.

W^t	D_i^t	$P(D_i^t \mid W^t)$	**Description**
i	1	p_w (0.95)	True Positive (Correct Detection)
i	0	$1 - p_w$ (0.05)	False Negative (Missed Detection)
$j \neq i$	1	p_c (0.05)	False Positive (Clutter)
$j \neq i$	0	$1 - p_c$ (0.95)	True Negative (Correct Non-Detection)

2.1.4 Cluster Graph and Inference

The topology of an HMM naturally forms an acyclic graph [2]. Due to this, the Bayesian Network (Figure 1) is converted to a Junction Tree cluster graph using the built-in functionality of the **emdw** library. This structure guarantees exact inference through a single forward-backward message-passing schedule, resulting in a balance between computational efficiency and inference accuracy. Due to the tree structure, the time complexity scales linearly, $O(T)$, with the number of time steps [2]. This cluster graph is shown in Figure 2 and the factors are assigned as follows:

- $P(W^t \mid W^{t-1})$ is assigned to a cluster with the scope $\{W^{t-1}, W^t\}$.
- $P(\mathbf{D}^t \mid W^t)$ is assigned to a cluster with scope $\{W^t\}$.

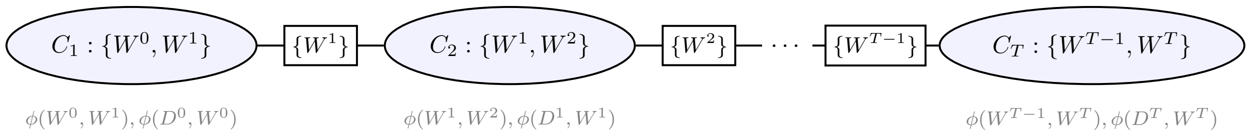


Figure 2: Cluster graph for exact inference.

For the trajectory inference task, Maximum A Posteriori (MAP) inference using the Max-Product algorithm was selected over Marginal inference with Belief Propagation (BP). This choice is motivated by the fact that MAP inference computes the most likely trajectory $(W^0, W^1, \ldots, W^T)$ which results in a path that is globally consistent and adheres to the wumpus's movement constraints as defined in the transition model. That is to say, the MAP estimates represent a valid movement of the wumpus through the grid world [3]. In contrast, Marginal inference using the BP algorithm identifies the most likely cell for the wumpus at each discrete time step independently [4]. This may lead to a sequence of locations that are infeasible if certain cells have high local probabilities.

2.2 Dataset 2

To evaluate the model's robustness, the model is extended to Dataset 2, which utilises a 20×20 grid. This significantly increases the number of possible wumpus locations to 400, meaning the transition factor expands into a 400×400 matrix, while each of the 400 binary detection factors per time step scales to a 2×400 conditional probability table. The parameters for this environment are defined as $p_w = 0.9$ and $p_c = 0.1$. This dataset serves as a critical test to verify that the implementation can handle larger grid sizes efficiently without a drastic computational cost. Despite the increased dimensionality, the Junction Tree structure remains. Consequently, the time complexity remains linear $O(T)$ and the computational resources (time or memory) scale proportionally with the size T.

2.3 Dataset 3

For Dataset 3, the grid environment expands into a 10×20 grid with $T = 20$ timesteps. Furthermore, the model parameters p_w and p_c are unknown. An appropriate method to determine the parameters is the Expectation-Maximization (EM) algorithm [5]. The EM algorithm is an iterative method to estimate the parameters, Θ, that maximise the data likelihood, $p(\mathcal{D}|\Theta)$, subject to the presence of latent variables. The objective of the E-step is to perform inference on the hidden variables given the current estimate of the parameters, Θ. This is done by calculating the posterior distribution, $p(\mathcal{H}|\mathcal{D}, \Theta)$, where $\mathcal{H}$ represents the latent variables. The responsibilities $\gamma_t(i)$ form the marginal distribution over every grid cells at each time t:

$$\gamma_t(i) \propto P(W_t = i) \cdot \underbrace{\left[p_w^{D_{t,i}}(1-p_w)^{1-D_{t,i}}\right] \prod_{j \neq i} \left[p_c^{D_{t,j}}(1-p_c)^{1-D_{t,j}}\right]}_{\text{Observation Likelihood } P(\mathbf{D}_t | W_t = i)} \tag{1}$$

The M-step aims to update the parameters to maximise the auxiliary function, $Q(\Theta)$. After some mathematical manipulations, the auxiliary function results in Equation 3.

$$\Theta^{\text{new}} = \arg\max_{\Theta} \mathcal{Q}(\Theta) \tag{2}$$

$$\mathcal{Q}(\Theta) = \sum_{t,i} \gamma_t(i) \left[D_{t,i} \log p_w + (1 - D_{t,i}) \log(1 - p_w) + \sum_{j \neq i} (D_{t,j} \log p_c + (1 - D_{t,j}) \log(1 - p_c)) \right] \tag{3}$$

To determine the update step for all model parameters, the partial derivative of Equation 3 is taken with respect to each parameter and set to zero. This results in the update equations shown in 4, where $S_t = \sum D_{t,i}$ is the total detection count across the entire 10×20 grid at time step t.

$$\hat{p}_w = \frac{1}{T}\sum_{t=0}^{T-1}\sum_{i=0}^{N-1}\gamma_t(i)D_{t,i}, \quad \hat{p}_c = \frac{\sum_{t=0}^{T-1}\sum_{i=0}^{N-1}\gamma_t(i)(S_t - D_{t,i})}{T(N-1)} \tag{4}$$

3 Code Implementation

3.1 Random Variables Definition

The model uses two primary vectors for RV management: `W_rvs` for the latent wumpus positions and `D_rvs` for observed detections. Storing the random variable IDs in these vectors allows efficient looping during the factor construction phase, as specific time dependent RVs (W^{T-1} and W^t) can be accessed directly via their vector indices. `D_rvs` is a 2D vector which indexes one RV for each time step and cell location.

```
vector<T> W_rvs(T_max);
vector<vector<T>> D_rvs(T_max, vector<T>(25));
```

Listing 1: Declaration of latent and observable random variable vectors.

3.2 Implementation of the Model Factors

To ensure that the tracking solution computes the Maximum A Posteriori (MAP) trajectory rather than the marginal probabilities, the factors were instantiated with specific operators. During the construction of both transition and detection factors, specialised pointers for marginalization (`DiscreteTable_MaxMarginalize`) and normalization (`DiscreteTable_MaxNormalize`) were passed to the `DiscreteTable` (`DT`) constructor. By defining the factors this way, the functions called by the cluster graph change their behavior: the `marginalize()` operation performs maximisation instead of summation, and the `normalize()` operation scales the factor such that its maximum value is 1.

3.2.1 Implementation of the Transition Factors

The transition model is implemented by iterating through each time step and constructing a transition factor $P(W^t \mid W^{t-1})$ that connects every current state to its previous. The implentation logic follows a few key steps. Firstly, within each iteration of the temporal loop, the unique IDs for the current and previous wumpus positions are retrieved from the `W_rvs` vector. Because the grid is represented using a 1D index (0 to 24), the code converts these indices back into (x, y) coordinates to facilitate the check for valid adjacent moves.

```
int x = i % width; // x-coordinate
int y = i / width; // y-coordinate
```

Listing 2: Conversion of grid indices to (x, y) coordinates

Coordinate offsets (dx and dy) are used to determine all potential subsequent wumpus location. To handle the grid boundaries, the proposed wumpus position is checked if it stays withing the grid dimensions. If a move is valid, it is assigned a probability of 0.25; and if a move would go out of bounds, the 0.25 probability is instead accumulated. This ensures that the probability of staying in a cell correctly reflects how many boundaries that the cell is touching (e.g., 0.5 in a corner).

```
// Define possible grid moves (up, down, left, right)
int dx[] = {0, 0, -1, 1};
int dy[] = {1, -1, 0, 0};
for (int move = 0; move < 4; move++) {
  int new_x = x + dx[move]; // wumpus new x-coordinate
  int new_y = y + dy[move]; // wumpus new y-coordinate
```

```
  if (new_x >= 0 && new_x < width && new_y >=0 && new_y < width) {
    // Valid transition
    int new_i = new_y * width + new_x; // Convert back to cell index
    transitionProbs[{i, new_i}] = 0.25;
  } else {
    // Invalid transition
    stay_prob += 0.25; // Accumulate probability of staying in cell
  }
}
transitionProbs[{i, i}] = stay_prob;// Prob of wumpus staying in the same cell
```

Listing 3: Transition factor logic.

A `std::map` is used to store only the non-zero probabililities to prevent creating a massive 25×25 table with many zeros which consequently improves computational performance. These sparse entries are then used to instantiate an `emdw DT` called `ptrTransition`, which is then stored in a vector of factor pointers for later use in the cluster graph.

```
transitionFactors.push_back(ptrTransition);
```

Listing 4: Vector of transition factor pointers

3.2.2 Implementation of the Detection Factors

The relationship between the latent wumpus position and the observed detection data $P(\mathbf{D}^t \mid W^t)$ is implemented through a nested looping structure that generates a detection factor for every cell at every time step. An outer loop is utilised to iterate through each time step t and an inner loop to iterate through every grid cell `loc`. This ensures that each cell in the grid world generates its own binary observation variable D^t_{loc} dependent on the wumpus's latent position W^t. For each cell, the probability of a detection (1) or no detection (0) is evaluated based on whether the wumpus is currently occupying that specific cell.

- $w\_pos == loc$ (True detection): The wumpus is in the cell. The probability of a correct detection is assigned the parameter p_w, while a missed detection is $1 - p_w$.
- $w\_pos \neq loc$ (Clutter measurement): The wumpus is not in the cell. The probability of a detection is p_c, and a correct non-detection is $1 - p_c$.

```
// loc is the location of the specific cell we are building a factor for
for (int loc = 0; loc < num_cells; loc++) {
  int D_curr = D_rvs[t][loc]; // Extract index of D RV at time t and grid location
  map<vector<T>, FProb> detectionProbs;
  // w_pos is the latent position of the wumpus
  for (int w_pos = 0; w_pos < num_cells; w_pos++){
    if (w_pos == loc) {
      // Detection location is equal to the wumpus position
      detectionProbs[{1, w_pos}] = pw; // Detection
      detectionProbs[{0, w_pos}] = 1.0 - pw;  // Missed detection
    } else {
      detectionProbs[{1, w_pos}] = pc; // Clutter
      detectionProbs[{0, w_pos}] = 1.0 - pc; // No detection
    }
  } // end of wumpus location loop
}
```

Listing 5: Detection factor initialisation.

The factor scope is defined as $\{D^t\_loc, W^t\}$ which is extracted from the W_rvs and D_rvs vectors. By using a `std::map` to store the `detectionProbs`, the sparse implementation remains efficient. These probabilities are used to instantiate DT factors which are then stored in a `detectionFactors` vector.

```
detectionFactors.push_back(ptrDetections);
```

Listing 6: Vector of detection factor pointers

3.3 Cluster Graph Construction

This stage involves assembling the defined factors into a cluster graph to perform inference. This process is handled using `emdw` functionality. Firstly, all factors are collected into a single vector of pointers (`factorPtrs`), and an observation map (`obsv`) is created to store all evidence of the observed RVs. The binary detection data is loaded from external dataset files and mapped to the corresponding D^t_{loc} variable IDs. This ensures that the inference procedure is conditioned on the actual data provided in the datasets. A `ClusterGraph` object is instantiated using the `BETHE` method which constructs a Junction Tree. The `loopyBP_CG` function executes exact inference on the tree structure. Once message passing has converged, the cluster graph contains the posterior information required to determine the most likely state of the wumpus at each time step.

```
ClusterGraph cg(ClusterGraph::BETHE, factorPtrs, obsv);
map<Idx2, rcptr<Factor> > msgs;
MessageQueue msgQ;
unsigned nMsgs = loopyBP_CG(cg, msgs, msgQ);
```

Listing 7: Cluster graph creation and message passing

3.4 Extraction of the MAP Estimates

The result of MAP message passing is a set of max-marginal beliefs. The most likely local combination of each max-marginal belief is consistent with the most likely global combination. For every time step t, the cluster graph is queried using `queryLBP_CG` for the belief of the latent variable W^t. This belief represents the maximum joint probability of the entire sequence passing through that specific state at time t. A nested loop then iterates through all possible grid cells to identify the cell that yields the highest value in the max-marginal factor. This identifies the specific position, at that time step t, that belongs to the overall most likely trajectory.

```
rcptr<Factor> beliefT = queryLBP_CG(cg, msgs, {W_rvs[t]})->normalize(); // Query
for (int row = 0; row < R; row++) {
  for (int col = 0; col < C; col++) {
    // Calculate the cell index and extract likelihood
    unsigned int cell_idx = row * C + col;
    double likelihood = beliefT->potentialAt({W_rvs[t]}, {(T)cell_idx});
    if (likelihood > max_likelihood) {
      max_likelihood = likelihood;
      best_cell = cell_idx;
    }
  }
}
```

Listing 8: Extraction of MAP estimates from max-marginal beliefs.

To facilitate the model evaluation phase, each inferred cell index is converted back into Cartesian (x, y) coordinates and exported to an external text file. Persisting the data in this manner ensures that the inferred trajectory remains structurally consistent with the ground-truth datasets. Once this process has been completed for every time step t, the inference procedure has identified the single most likely sequence of locations for the wumpus across all timesteps.

3.5 Scalability and Implementation: Dataset 2

Transitioning the implementation to Dataset 2 requires minimal changes due to the initial software design of Dataset 1. The model parameters were updated to accomodate the larger environment, including the number of time steps ($T_{max} = 20$), grid dimensions ($num\_cells = 400$), and detection probabilities ($p_w = 0.90, p_c = 0.1$). The `locationDom` was expanded by iterating over 400 indices to accommodate the increased number of states. Importantly, the logic for factor construction, cluster graph creation, and MAP inference remained identical to the Dataset 1 implementation, demonstrating the scalability of the model. Additionally, the evidence-loading function was refined with zero-padding logic to handle varying filenames (e.g., `data_file001.txt` vs `data_file010.txt`), ensuring robust data ingestion for the extended timeline.

3.6 Dataset 3: EM Implementation and Parameter Estimation

The implementation of the EM algorithm to estimate the unknown detection parameters p_w and p_c for Dataset 3 is structured around an interative convergence loop that alternates between inference and parameter refinement [5]. Firstly, the parameters are initialised intuitively with the values $p_w = 0.8$ and $p_c = 0.1$, representing a guess of high detection accuracy and low noise. In the E-step, `emdw` functionality performs belief propagation using the message-passing algorithm with `SumMarginalize` operators to calculate the responsibilities $\gamma_t(i)$ for each state, as defined in Equation 1. $\gamma_t(i)$ is a soft assignment that determines how likely it is that the Wumpus is in cell i at time t. A key efficiency in the implementation was constructing the transition factors once outside the loop, as they do not depend on the model parameters, and simply adding them to the `factorPtr` during each EM iteration.

The M-step updates p_w and p_c based on Equations 4 using the computed E-step beliefs. Specifically, the numerator for p_w is calculated as the sum of beliefs at locations where a detection actually occurred, while the numerator for p_c represents the weighted sum of false detections (all detections in the grid excluding the wumpus's likely location).

```
for (int i = 0; i < num_cells; i++) {
  double gamma_ti = beliefT->potentialAt({W_rvs[t]}, {(T)i});
  int D_ti = ((int)obsv[D_rvs[t][i]] == 1) ? 1 : 0;
  if (D_ti) pw_num += gamma_ti;
    pc_num += gamma_ti * (St - D_ti);
  }
pw = pw_num / T_max; // Updated pw
pc = pc_num / (T_max * (num_cells - 1)); // Updated pc
```

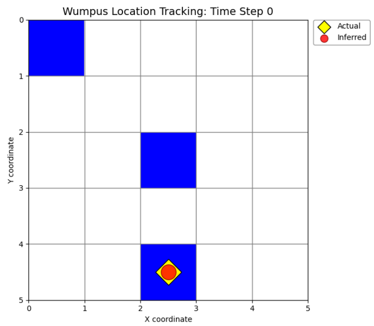
Listing 9: M-step: Parameter update step.

It is critical to note that Belief Propogation is used to determine the marginals during the EM phase rather than max-marginals. Only once the parameters converged is the MAP approach applied to the final cluster graph (with estimated p_w and p_c parameters) to extract the globally consistent wumpus trajectory.

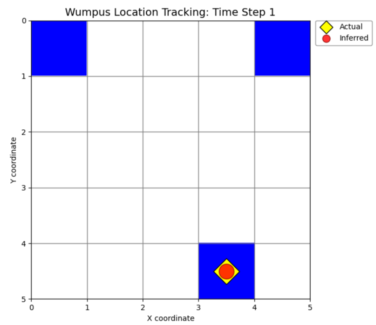
4 Analysis of Results

Model performance is qualitatively assessed using a Python script to visualise the wumpus's path on the grid world. The script uses command-line arguments to switch between datasets, automatically adjusting the grid dimensions and file paths for Datasets 1, 2, and 3. The detection data is overlayed with the inferred trajectory and ground-truth data, allowing for a clear visual comparison of the results. Figure 3 displays these results for Dataset 1 at various time steps, where the yellow star represents the ground-truth wumpus location and the red dot represents the location inferred by the MAP algorithm. The corresponding visualisation for Dataset 2 and 3 is shown in Figure 4 and 5, respectively. While investigating the qualitative results of Dataset 3, a deviation from the ground truth for the first few time steps is revealed. These errors suggest that the model's initial uncertainty is only resolved once the posterior probability peaks more sharply around the actual coordinates, leading to the consistent tracking seen in the last half of the simulation.

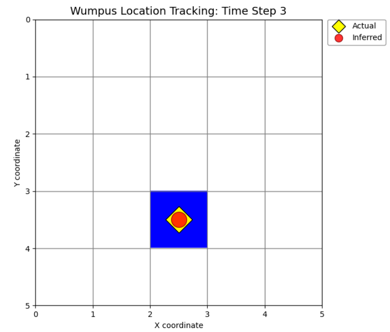
A quantitative evaluation is used to objectively analyse the performance and scalability of the models. The primary metrics utilised include: Accuracy, Neighbourhood Accuracy, Root Mean Square Error and



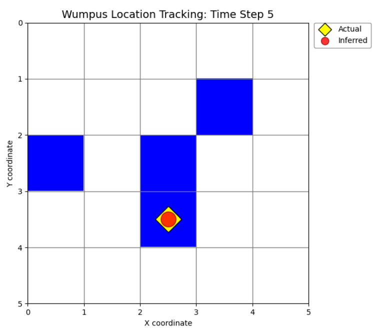
(a) Time $t = 0$



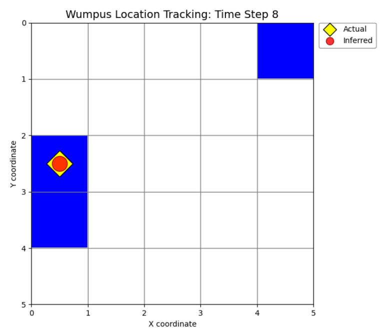
(b) Time $t = 1$



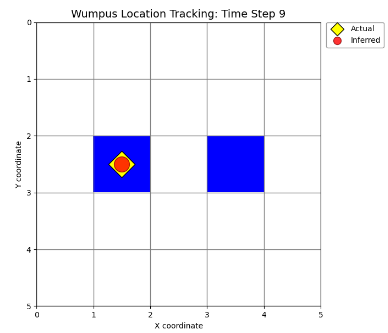
(c) Time $t = 3$



(d) Time $t = 5$

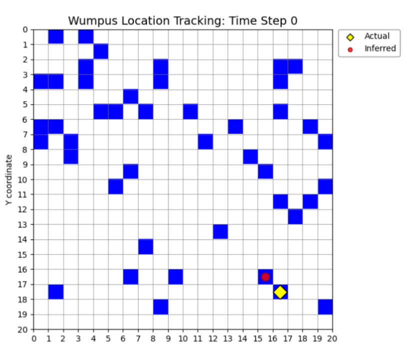


(e) Time $t = 8$

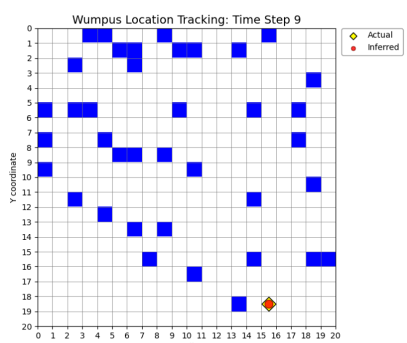


(f) Time $t = 9$

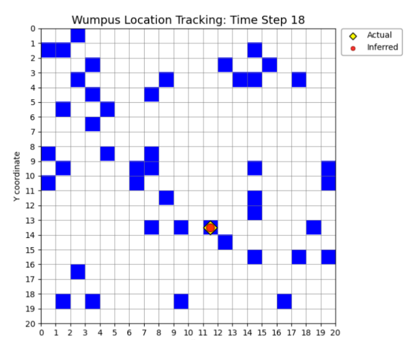
Figure 3: Dataset 1 tracking performance across select time steps.



(a) Time $t = 0$

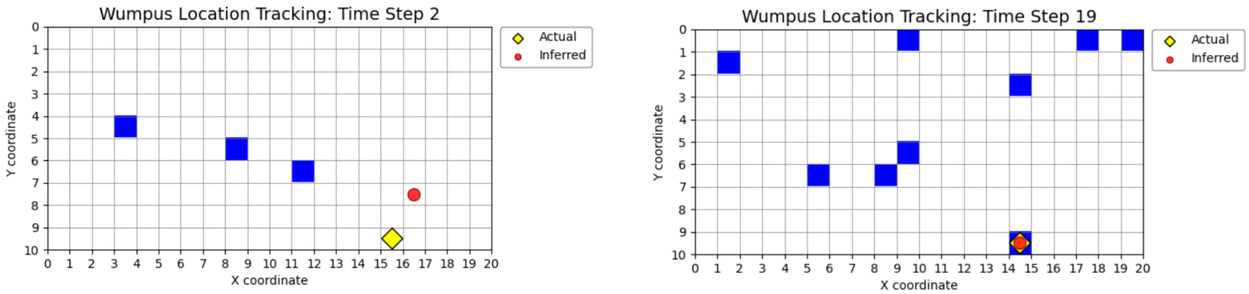


(b) Time $t = 9$



(c) Time $t = 18$

Figure 4: Dataset 2 tracking performance on the 20×20 grid environment.



(a) Time Step 2

(b) Time Step 19

Figure 5: Dataset 2 tracking performance on the 10×20 grid environment.

Mean Manhattan Distance (MMD). Accuracy represents the percentage of times steps where the inferred MAP location W^t matches the ground-truth coordinates. To quantify classification deviations, the RMSE was calculated to measure the average Euclidean distance between the predicted and actual positions.

$$RMSE = \sqrt{\frac{1}{T}\sum_{t=0}^{T-1}\left[(x_t - \hat{x}_t)^2 + (y_t - \hat{y}_t)^2\right]} \tag{5}$$

Neighborhood Accuracy and MMD obtain a more naunced understanding of the model's tracking ability, especially in the more complex environment of Dataset 2. The Neighborhood Accuracy measures the percentage of time steps that infer the location within one grid step of the actual position. Since the classes represent a physical grid, an inference that is one cell removed from the ground truth is more valuable than one on the other side of the map. Lastly, the MMD provides an intuitive error measurement that represents the average number of moves separating the MAP estimate from the wumpus's true location.

$$MMD = \frac{1}{T}\sum_{t=0}^{T-1}\left(|x_t - \hat{x}_t| + |y_t - \hat{y}_t|\right) \tag{6}$$

Table 3: Quantitative Results for Wumpus Tracking (Datasets 1, 2, & 3).

Dataset	**Acc. (%)**	**N-Acc. (%)**	**RMSE**	**MMD**	**Remarks**
Dataset 1	100.00%	100.00%	0.0000	0.00	Perfect tracking.
Dataset 2	85.00%	85.00%	0.5477	0.30	Deviations at $t \in \{0, 5, 11\}$.
Dataset 3	75.00%	80.00%	1.0488	0.60	Deviations at $t \in \{0, 1, 2, 3, 6\}$.

**Acc: Exact Match Accuracy; N-Acc: Neighborhood Accuracy (within 1 cell);*
RMSE: Root Mean Square Error; MMD: Mean Manhattan Distance (steps).

As shown in Table 3, the model achieved a perfect success rate of 100% on Dataset 1, indicating that the MAP inference successfully filtered out detection inaccuricies in the smaller 5×5 grid environment. For the more complex 20×20 environment in Dataset 2, the accuracy decreased to 85%, with an RMSE of 0.5477 units. This performance drop is expected given the significantly larger grid world (400 cells) and the higher likelihood of clutter measurements p_c. In Dataset 2, an MMD of 0.30 steps suggests that across the 20 time steps, the model was, on average, less than one-third of a grid step away from the true wumpus location. This confirms that the trajectory remains highly reliable despite the increased state space of 400 cells. The low RMSE along with the MMD confirms that even when the model incorrectly classified the wumpus's location, the prediction still remained in the spatial proximity of the ground truth.

While Dataset 3 presents the most challenging tracking scenario, the model still achieves a 75.00% accuracy. The Neighborhood Accuracy of 80.00% suggests that even when the model misses the exact cell, it remains within one unit of the true location the majority of the time. This is further confirmed by the

RMSE of 1.0488 and MMD of 0.60, indicating that deviations are generally small and localised. Finally, in evaluating the Dataset 3 parameter estimation, it is observed that the EM algorithm determined the unknown sensor parameters, with p_w and p_c converging to 0.8154 and 0.0761, respectively, within 7 iterations. Thus, p_w increased from the initial guess ($0.8 \rightarrow 0.815$), while p_c decreased ($0.1 \rightarrow 0.076$). This confirms that the EM algorithm learned that the wumpus detections are slightly more accurate and less noisy than originally assumed.

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